CN 202455480 U — English machine translation English machine translation of CN 202455480 U retrieved through EPO Espacenet (machine translation displayed with Espacenet's own notice: "The wording below is an initial machine translation of the original publication") on 2026-08-08 from https://worldwide.espacenet.com/patent/search/family/046871301/publication/CN202455480U. The original Chinese-language publication is stored separately and is intended to accompany any IDS submission if selected by counsel. Text extracted programmatically and reformatted for legibility without editorial changes; the Espacenet UI header lines were removed. This document is not represented as a certified or authoritative translation. Publication CN 202455480 U · 2012-09-26 Title (EN) Digital watermark system for verifying digital television copyright Applicant Chengdu Doyen Information Technology Co., Ltd. Inventors Zejun Zeng; Shengwu Xu; Keyu Gong; Qipeng Yi; Wei Huang Priority / application CN 201120556629 U · 2011-12-28 IPC H04N 21/4367; H04N 21/8358 Abstract (English, as displayed by Espacenet) The utility model discloses a digital watermark system for verifying a digital television copyright. The digital watermark system comprises a program operator side, a satellite network, a wired front end, a wired network, a user side, a digital watermark satellite receiver and a watermark monitoring side, wherein the digital watermark satellite receiver is used for receiving a satellite signal transmitted by the satellite network, decoding the satellite signal into a video signal, embedding a digital watermark comprising the ID (Identity) of a smart card of the satellite receiver into the video signal, and transmitting the video signal comprising the digital watermark to the wired front end; and the watermark monitoring side is used for receiving and monitoring digital television program contents from the wired network, and making an alarm if digital watermarks implied in the digital program contents are illegal. Description (English machine translation) Technical Field The utility model belongs to the field of digital television, and in particular relates to a system for verifying the copyright of digital television programs by using digital watermarks. Background Art Digital TV is increasingly accepted and loved by the majority of users due to its controllable characteristics. However, it is precisely because of the storable and editable characteristics of digital TV that the piracy of digital TV programs has become increasingly serious. The copyright protection of digital TV program content is urgent. How to effectively manage and control the content of digital TV programs has become the focus of attention of program operators. Program operators transmit encrypted programs to satellite receiving devices of various local stations via satellite. The satellite receiving equipment decrypts the encrypted programs into transparent programs and then broadcasts them on their respective cable networks. In order to avoid paying viewing fees to program operators, some network operators may steal a program from other cable networks and add it to their own content for playback. In order to alleviate this problem, the copyright control of existing programs is mainly protected by CA (Conditional Access) conditional reception technology, that is, scrambled and encrypted content is broadcast, and each user terminal can only receive and decrypt specific content. Fingerprint display technology is mainly used for copyright identification. However, since fingerprint display technology requires program operators to send fingerprint information to a national network or a regional network, the fingerprint information can be seen in the national or regional network, resulting in poor experience for other users, and it is difficult to trace the network operator, so the actual anti- piracy effect is not good. Utility Model Content In view of the above problems, the purpose of the utility model is to provide a system for verifying the copyright of digital television programs by using the digital watermark of the network operator, so as to achieve the effect of quickly locating the source of piracy without affecting the viewing of users on other networks. The above-mentioned purpose of the utility model is achieved through the following technical solutions: a digital watermark system for digital television copyright verification, including a program operator end, a satellite network, a cable front end, a cable network and a user end, and also including a digital watermark satellite receiver and a watermark monitoring end; the digital watermark satellite receiver is used to receive a satellite signal transmitted from the satellite network, and decrypt the satellite signal into a video signal, and at the same time embed a digital watermark containing the smart card ID of the satellite receiver into the video signal, and finally send the video signal containing the digital watermark to the cable front end; the watermark monitoring end is used to receive and monitor the digital television program content from the cable network, and if it is found that the digital watermark implicit in the digital program content is illegal, an alarm is issued. Furthermore, the digital watermark is semi-transparent and will not affect the viewing of users. The beneficial effects of the utility model are as follows: a digital watermark containing the smart card ID of the satellite receiver is embedded into a video signal through a digital watermark satellite receiver, and then a watermark monitoring terminal monitors the digital watermark of the current digital TV program content. If the digital watermark is found to be illegal, an alarm is issued, so that the program operator can quickly locate the source of piracy based on the digital watermark to achieve the purpose of copyright verification; at the same time, a digital watermark is only transmitted in a wired network and will not affect other users. BRIEF DESCRIPTION OF THE DRAWINGS FIG1 is a schematic diagram of the module structure of a specific embodiment of the utility model. DETAILED DESCRIPTION All features disclosed in this specification, or steps in all methods or processes disclosed, except mutually exclusive features and/or steps, can be combined in any manner. Any feature disclosed in this specification (including any claims, abstracts and drawings), unless otherwise stated, may be replaced by other alternative features that are equivalent or have similar purposes. That is, unless otherwise stated, each feature is only an example of a series of equivalent or similar features. At the same time, the description of alternative features in this specification is a description of equivalent technical features and shall not be regarded as a contribution to the public. If terms in this specification (including any claims, abstract and drawings) have both general meanings and meanings specific to the art, they shall be defined as the meanings specific to the art unless otherwise specified. The module structure diagram of a specific embodiment of the utility model is shown in Figure 1. The utility model is composed of a program operator end, a satellite network, a digital watermark satellite receiver, a cable front end, a cable network, a user end and a watermark monitoring end. The program operator provides digital TV program content, which is then broadcasted through a satellite network; the digital watermark satellite receiver receives satellite signals from the satellite network, decrypts the satellite signals into video signals, and embeds the digital watermark containing the satellite receiver's smart card ID into the video signal, and finally sends the video signal containing the digital watermark to the wired front end; the wired front end sends the video signal containing the digital watermark to each user end through the wired network; the watermark monitoring end receives and monitors the digital TV program content from the wired network, and issues an alarm if it finds that the digital watermark hidden in the digital program content is illegal. In this way, after the watermark monitoring end issues an alarm, the program operator can read the smart card ID through the digital watermark. Since a smart card ID has its specific network operator, the program operator can quickly locate the network operator where piracy occurs. The program operator can notify the network operator of the copyright piracy and require the network operator to disconnect the connection with the pirated cable network, thereby pursuing the infringement of the network operator and the pirated cable network to achieve the purpose of protecting copyright. At the same time, since different digital watermarks are only for the network behind a satellite receiver, it will not affect users of other networks. According to an embodiment of the utility model, the digital watermark is semi-transparent, which can make users experience a better viewing experience. Claims (English machine translation) 1. A digital watermark system for digital television copyright verification, comprising a program operator end, a satellite network, a cable front end, a cable network and a user end, characterized in that: It also includes a digital watermark satellite receiver and a watermark monitoring terminal; The digital watermark satellite receiver is used to receive satellite signals from a satellite network, decode the satellite signals into video signals, embed the digital watermark containing the satellite receiver smart card ID into the video signals, and finally send the video signals containing the digital watermark to the wired front end; The watermark monitoring terminal is used to receive and monitor the digital television program content from the cable network, and if it is found that the digital watermark hidden in the digital program content is illegal, an alarm is issued. 2. The digital watermark system for digital television copyright verification according to claim 1, characterized in that: The digital watermark is semi-transparent.